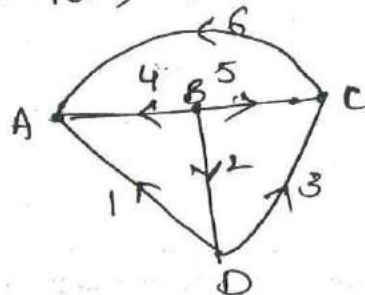


GRAPH THEORY

- * For a given graph augmented incidence matrix is unique.
- * All the information regarding the graph can be represent mathematically in concise form is called as "Augmented Incidence Matrix".



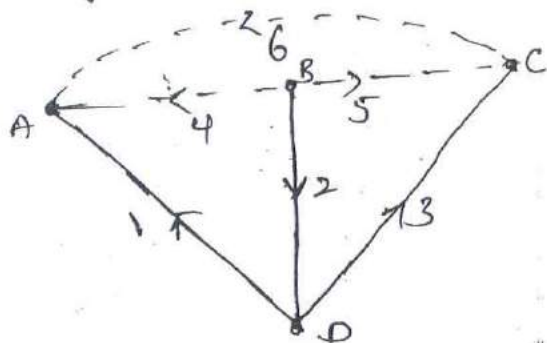
- Tree
- co-Tree
- Tree Branch (Twig)
- Link (chord)

Tree : is a connected subgraph, in which there is no closed path. (1, 2, 3 forms a tree).

co-tree :- Complementary Tree. (4, 5, 6) forms a co-tree.

tree branch :- the branch which forms a tree is called "tree branch" it is indicated by thick line.

- Link:- The branch which is disconnected from tree is called "link".
- The combination of links will form a co-tree. The link is indicated by dotted line.



Tree:- tree is a connected subgraph, it connects all nodes of the graph, but it does not consist of any

$$\text{Tree} = \{1, 2, 3\}$$

co-tree:- the set of connections which are disconnected to form a tree is called as "co-tree" (complementary tree).

$$\text{co-tree} = \{4, 5, 6\}$$

Treebranch:- the branch which form a tree is called as "treebranch" (Twig).

$$\text{Total NO: of Tree branches} = N - 1$$

where $N = \text{no: of nodes}$ ***

Link:- The branch which is disconnected to form a tree is called as "link" (chord).

$$\text{Total NO: of links} = b - N + 1$$

where $b = \text{no: of branches of N/w (not graph)}$
 $N = \text{no: of nodes}$ ***

Note:-

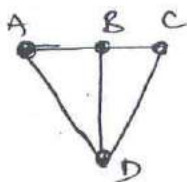
If the N/w is unknown then assume that both N/w of graph having same branches.

$$\text{Total NO: of Trees possible for } N \text{ nodes} = N^{N-2}$$

or N/w ***

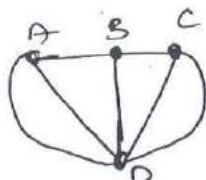
This formula having some limitations:

- 1) direct connection should be present b/w all the nodes.
- 2) no repeated branch is present b/w two nodes.



$$N=4$$

$$\text{trees} = 4 - 2 = 16$$



$$N=4$$

$$\text{trees} = 4 - 2 = 16$$

* for both the graph the no. of trees possible is same.

So we can't use the above formula for these two graphs.

Since for first graph there is no direct connection b/w A & C.

And for the second graph repeated branch will present b/w two nodes.

Total NO. of Trees possible for any graph

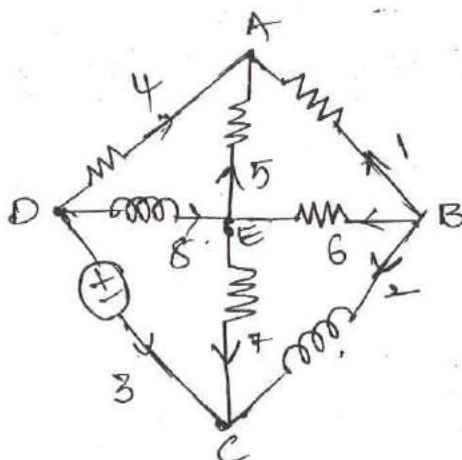
$$= |A A^T|$$

General expression for any graph

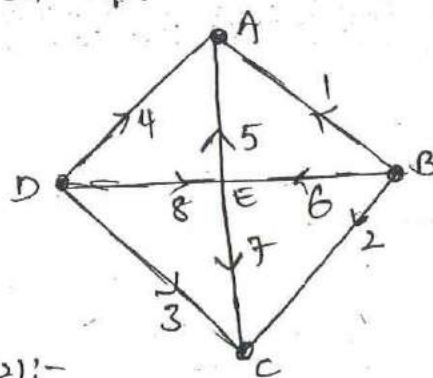
where A = reduced incidence matrix

* Tieset Matrix:-

Develop Tieset Matrix for the N/W shown.

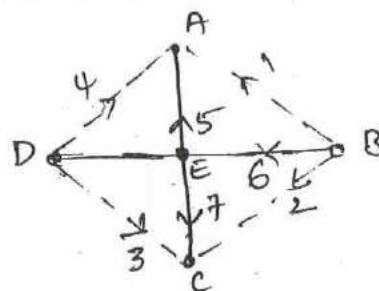


Step (1):- Develop a graph for given N/W.



Step (2):-

Develop a tree for a graph.



Step (3):-

i) Identify total NO. of Basic loops (or) fundamental loops (or) F-loops (or) Independent loops.

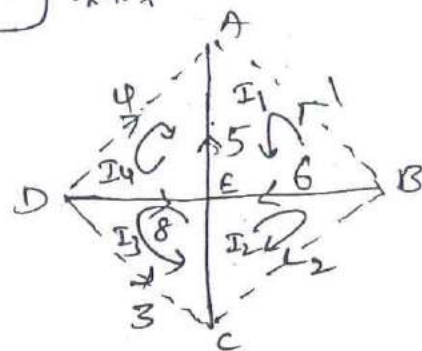
ii) Basic loop means it should consist of only one link.

iii) So we can conclude that total NO. of Basic loops is equal to total NO. of links.

iv) Basic loop direction is same as that of link direction.
 (Current =)

$$\text{Total No. of Basic Loops} = b - n + 1 \quad ***$$

	V_1	V_2	V_3	V_4	V_5	V_6	V_7	V_8	
	1	2	3	4	5	6	7	8	
$[C] = [A]$					-1	-1			$\leftarrow I_1$
		+1				-1	-1		$\leftarrow I_2$
			+1				-1	-1	$\leftarrow I_3$
				+1	-1			-1	$\leftarrow I_4$
	Unit Matrix (U)				CB - tree branches (B)				



* There is no standard name for Cutset & Tree set matrices. only present for Incident Matrix (A).

* for closed loops we write KVL not KCL. so we take voltages.

KVL:

$$V_1 - V_5 - V_6 = 0$$

$$V_2 - V_6 - V_7 = 0$$

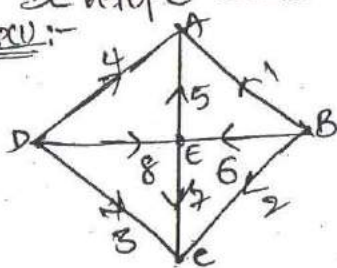
$$V_3 - V_7 - V_8 = 0$$

$$V_4 - V_5 - V_8 = 0$$

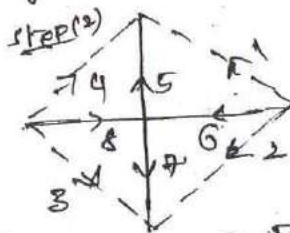
Now the entire matrix can be divided into two parts. i.e. U & CB as shown in the above figure.

$$[C] = [U; C_B] \quad * \text{this splitting is useful to identify Cutset matrix for a given tree matrix.}$$

Sol: Step (1): Develop cutset matrix for the N/W shown (for the above Pblm).
Develop the graph for the given N/W.



$\Rightarrow$ Tree



Step (2): Develop a tree for a graph.

Step (3): - (i) Identify total No. of Basic cutsets (or) fundamental cutsets (or) f-Cutsets.

ii) When ever you are cutting a set of elements only one ^{tree} branch should present in Basic cutset should consist of only one tree branch.

